

Communication: Language: easy to understand, answers question, follows instructions, concise, precise Chemistry: proper vocab, notation, spacing, logical Math: organized, units, sig digs, math notation, concluding statement, spacing	Overall: vague/basic/OK/good/clearly written
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SCH4U

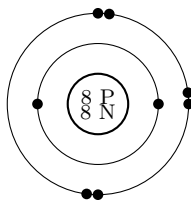
Name: _____

Monday, March 1, 2025

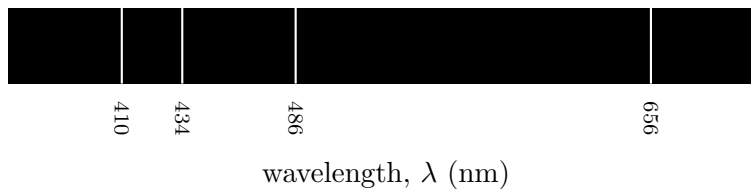
Unit 1: Structures & Properties Test

Total: /35	Comm: /10
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1. [2 marks] What is wrong with the following Bohr-Rutherford diagram with regards to Quantum Mechanics? Be specific.



2. [3 marks] Below is the emission spectra for hydrogen when an electron falls to $n = 2$. Where would an emission line for hydrogen be if an electron falls from $n = 5$ to $n = 3$? Indicate the location of this new emission line relative to the other emission lines. Is this energy transition high energy or low energy? Explain with or without a diagram. Calculations are not needed to answer this question.



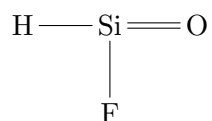
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6. [3 marks] Draw the ELD (energy level diagram) for neutral ground state As (element 33). Label the subshells.

7. [3 marks] Draw the most stable Lewis structure of XeF_4^{2+} . Verify your answer using formal charges.

8. [2 marks] Why is anomalous electron configuration not the same as hybridization? Provide an example.

9. [4 marks] Draw the orbital overlap showing the bonding in the following structure. Use different colours for each bond (if you have them). Label each bond as being σ or π .



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10. [2 marks] What is the same & what is different between the following: AB_2E & AB_2E_2 ? Be precise.
11. [2 marks] Why are ionic solids brittle? What is unique about the structure that allows it to do so? Provide an example of an ionic solid.
12. [2 marks] Explain the humour behind the meme. Be sure to explain/describe the theory needed in order to fully appreciate the chemistry humour in the meme.

